Jian Weng

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Education

University of California, Los Angeles **Los Angeles, CA**
6th year Ph.D Candidate, Advisor: Tony Nowatzki *Aug. 2017 – Present*

Shanghai Jiao Tong University **Shanghai, China**
B.Eng. of Computer Science, ACM Honored Class *May 2013 – July 2017*
Thesis: Compiling Domain Specific Language to Reconfigurable Accelerators

Conference Publications

- OverGen: Improving FPGA Usability through Domain-specific Overlay Generation.
  Sihao Liu=, **Jian Weng=**, Dylan Kupsh, Atefeh Sohrabizadeh, Zhengrong Wang, Licheng Guo, Jiuyang Liu, Maxim Zhulin, Lucheng Zhang, Jason Cong, Tony Nowatzki.
  International Symposium on Microarchitecture (MICRO), 2022, **Best Paper Runner-up**
- Near-Stream Computing: General and Transparent Near-Cache Acceleration.
  Zhengrong Wang, **Jian Weng**, Sihao Liu, Tony Nowatzki.
  International Symposium on High-Performance Computer Architecture (HPCA), 2022
- UNIT: Unifying Tensorized Instruction Compilation.
  Jian Weng, Animesh Jain, Jie Wang, Leyuan Wang, Yida Wang, and Tony Nowatzki.
  International Symposium on Code Generation and Optimization (CGO), 2021
- Stream Floating: Enabling Proactive and Decentralized Cache Optimizations.
  Zhengrong Wang, **Jian Weng**, Jason Lowe-Power, Jayesh Gaur, Tony Nowatzki. *International Symposium on High-Performance Computer Architecture (HPCA), 2021*, **Best Paper Runner-up**
- DSAGEN: Synthesizing Programmable Spatial Accelerators.
  Jian Weng=, Sihao Liu=, Vidushi Dadu , Zhengrong Wang , Preyas Shah, and Tony Nowatzki.
  International Symposium on Computer Architecture (ISCA), 2020, **IEEE Micro Honorable Mentions**
- A Hybrid Systolic-Dataflow Architecture for Inductive Matrix Algorithms.
  Jian Weng, Sihao Liu, Zhengrong Wang, Vidushi Dadu, and Tony Nowatzki.
  International Symposium on High-Performance Computer Architecture (HPCA), 2020
- Towards General Purpose Acceleration by Exploiting Common Data-Dependence Forms.
  Vidushi Dadu, **Jian Weng**, Sihao Liu, and Tony Nowatzki.
  International Symposium on Microarchitecture (MICRO), 2019 **IEEE Micro Top Picks**
- Hybrid Optimization/Heuristic Instruction Scheduling for Programmable Accelerator Codesign.
  Tony Nowatzki, Newsha Ardalani, Karthikeyan Sankaralingam, and **Jian Weng**.
  Parallel Architectures and Compilation Techniques (PACT), 2018

Journal Publications

- Unifying Spatial Accelerator Compilation with Idiomatic and Modular Transformations.
  Jian Weng, Sihao Liu, Dylan Kupsh, Tony Nowatzki.
  IEEE Micro Special Issue on Compiling for Accelerators, 2022
- DAEGEN: A Modular Compiler for Exploring Decoupled Access Execte Accelerators.
  Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
  Computer Architecture Letters (CAL), 2019

Workshops and Tutorials

- OverGen: Decoupled-Spatial Architecture Framework
  Sihao Liu, **Jian Weng**, Dylan Kupsh, and Tony Nowatzki.
  International Symposium on Microarchitecture (@MICRO'22)

- A Full-Stack Infrastructure for Automating Spatial Architecture Research
  Jian Weng, Sihao Liu, Dylan Kupsh, and Tony Nowatzki.
  Workshop on Democratizing Domain-Specific Accelerators (@MICRO'22)
- Generality is the Key Dimension in Accelerator Design
  Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
  Workshop on Languages, Tools, and Techniques for Accelerator Design (@ASPLOS'21)
- DSAGEN: Democratizing Decoupled Spatial Architecture Research
  Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
  International Symposium on Microarchitecture (@MICRO'20)

Invited Talks

Synthesizing Programmable Accelerators: A Full-Stack Perspective
- University of Illinois, Urbana Champaign, ILLIXR Consortium — Nov. 9th, 2022
- Institute of Computing Technology, Chinese Acadamy of Sciences — Nov. 19th, 2022
- Georgia Institute of Technology, EIC Lab — Nov. 23rd, 2022

Synthesizing Programmable Accelerators: A Compiler Perspective
- University of Michigan, Computer Engineering Lab — Feb. 23rd, 2022
- University of Illinois, Urbana Champaign, CAP Seminar — Feb. 8th, 2022
- University of Washington, SAMPL Research Group — Jan. 27th, 2022

Open-Source Projects and Infrastructures

Democratizing Spatial Accelerator Design
Coauthor — *Aug. 2017 – Now*
- Develop the software stack for spatial architecture.

TVM — **Apache**
Committer — *Aug. 2017 – Now*
- Implement the `tvm.te.hybrid` module to enhance the expressiveness of the framework.
- Write tutorials on performance tuning and customized compilation pass.

Professional Experiences

Amazon Web Service — **East Palo Alto**
Applied Scientist Intern, Mentor: Yida Wang — *June 2019 – Sep. 2019, Jan. 2020 – Sep. 2020*
- Compile the newly emerging tensorized instructions to diverse hardware platforms (CGO 2021).
- Automatically integrate cuDNN to a next-generation machine learning framework .

Selected Awards

MICRO 2022 Best Paper Runner-up	*Oct. 2022*
UCLA Disseration Year Fellowship	*Sep. 2022*
QualComm Fellowship Finalist	*Aug. 2022*
Facebook Fellowship Finalist	*April 2022*
UCLA Graduate Student Fellowship	*Nov. 2017, Mar. 2021*
IEEE Micro Top Picks Honorable Mention	*2021*
HPCA Best Paper Runner-up	*2021*
IEEE Micro Top Picks	*2020*